Few-Shot-RAG Ontology-Driven Prompting for Ontology Alignment with LogMapLLM

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Abstract

Interactive ontology matching escalates uncertain correspondences to an oracle, for which large language models (LLMs) remain viable candidates. Building on our prior work, LLMs-as-Oracles with LogMapLLM, we introduce *few-shot-RAG ontology-driven prompting*. The approach retrieves positive example correspondences from LogMap's anchors and pairs them with sibling-swapped negatives for in-context learning. Because anchors are produced by the matcher itself, the method is self-supervised and does not require manually curated examples. We evaluate over three OAEI tracks using eight LLMs. Where oracles fail conservatively, retrieval yields substantial sensitivity-driven gains; for sensitivity-saturated oracles the benefit is smaller and can reverse as the number of examples grows.

Keywords

Ontology Matching, Large Language Models, Ontology Driven Prompting, In-context Learning

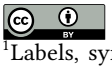
1. Introduction

Ontology matching is crucial for semantic interoperability, yet matchers often rely on lexical and structural heuristics, leading to ambiguous correspondences that cannot be easily resolved. Interactive matching escalates such cases to an oracle [1], an authoritative source that confirms or rejects each candidate. Often, the oracle is a human expert, whose time and attention is both scarce and costly. Large language models (LLMs) offer a viable, higher-throughput alternative, as demonstrated in our prior work, where we effectively used LLMs as oracles [2, 3] with ontology-derived prompts[1] for interactive matching in LogMap [4]. While that work addressed prompt-enrichment with *descriptive context*, it did not, however, explore *demonstrative* context. That is, showing task-specific examples to the LLM oracle to more appropriately condition its response, also known as in-context learning [5].

To bridge that gap, we investigate *few-shot-RAG ontology-driven prompting*. For each escalated correspondence, a retriever fetches relevant labelled examples from LogMap's anchors [3] (i.e., high-confidence correspondences produced during the matching process) and inserts them into the prompt as few-shot example demonstrations. Half of the retrieved examples are kept as positives, and half are corrupted to negatives by sibling substitution. Since the anchors (i.e., reliable mappings) are produced by the matcher itself, the method remains self-supervised, requiring no additional input besides the ontologies themselves and an oracle response. Thus, the system is deployable in environments where no known (partial) reference alignment exists.

We evaluate our approach over three OAEI[2] tracks [6]. Those being Anatomy, i.e., HUMAN–MOUSE [7], Bio-ML [8], and Knowledge Graph [9]. For our LLM oracle model selection, we aim to capture a balance between capability, cost, and openness; selecting primarily for open-weight models that allow for

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[1]Labels, synonyms, and a bounded structural neighbourhood relating to entities involved in the alignment for prompt enrichment [2, 3].
[2]Ontology Alignment Evaluation Initiative, https://oaei.ontologymatching.org/

reproducibility within a controlled setting. Preliminary results suggest that the benefit of including maximally relevant anchor-derived examples within the prompt is strongly task-dependent. In particular, observed gains are marginal where the zero-shot oracle is already highly sensitive, but substantial on Knowledge Graph matching, where conservatism appears to be the dominant failure mode.

2. Preliminaries

Ontology matching. Ontology matching is the task of finding a set of *correspondences*, referred to as an *alignment*, denoted $\mathcal{M}$, between two or more ontologies, such that entities from one ontology *correspond to* those in another [10]. A *correspondence* between a *source ontology* $\mathcal{O}_{src}$ and *target ontology* $\mathcal{O}_{tgt}$ is defined as a tuple $\langle e_{src}, e_{tgt}, r, c\rangle$ where e_{src} is drawn from $\mathrm{Sig}(\mathcal{O}_{src})$ and e_{tgt} from $\mathrm{Sig}(\mathcal{O}_{tgt})$; r is a relation in $\{\equiv, \sqsubseteq, \sqsupseteq\}$, and c is a confidence value in $[0, 1]$. In this work, we fix r to $\equiv$ and write $\langle e_{src}, e_{tgt}\rangle$ to mean an equivalence correspondence.

Interactive matching and oracles. An automatic matcher must resolve uncertain correspondences that do not evidently hold. *Interactive matching* escalates such cases to an authority (e.g., *a human-in-the-loop*), known as an oracle [1]. Let $\mathcal{M}_{ask}$ denote the set of *mappings to ask*, i.e., the subset of correspondences that an automatic matcher has generated but cannot fully commit to without further evidence. An *oracle* is a function $\mathrm{Or}\colon \mathcal{M}_{ask} \to \{\mathsf{true}, \mathsf{false}\}$ which the matcher consults, accepting the correspondences the oracle confirms and discarding the rest. By adopting LogMap [4] as the matcher, its interactive mode can be used to escalate cases where its lexical and structural evidence is insufficient.

Evaluation metrics. Oracle quality is measured on $\mathcal{M}_{ask}$ against a reference alignment by *sensitivity* (true-positive rate), *specificity* (true-negative rate), and *Youden's index* $J = \text{sensitivity} + \text{specificity} - 1$, which summarises discriminative ability independently of class balance. The final alignment is evaluated using precision, recall, and their harmonic mean F_1, reported as secondary.

3. Ontology-Driven Prompting for LLM Oracles

A language model relies on parametric knowledge and in-context learning [5] to effectively judge any $\langle e_{src}, e_{tgt}\rangle \in \mathcal{M}_{ask}$. *Ontology-driven prompting* assembles each oracle query using context from the ontologies themselves. For any correspondence in $\mathcal{M}_{ask}$, each of its entities act as input to a context function $\mathrm{ctx}(\cdot)$, which outputs related textual annotations and structural context. The oracle query is assembled as a task instruction, an optional few-shot slot (which can be empty, as in [2]), the output from $\mathrm{ctx}(e_{src})$ and $\mathrm{ctx}(e_{tgt})$, and the question of whether r holds between the two entities (Figure 1). The LLM oracle response is parsed as $\{\mathsf{true}, \mathsf{false}\}$ and returned to the matcher as the value $\mathrm{Or}(\langle e_{src}, e_{tgt}\rangle)$. The design of $\mathrm{ctx}(\cdot)$ must consider how much of the ontology's contextual information to include. Too little leaves the model guessing from a label, while too much dilutes discriminative evidence and inflates the token cost of every query.

Few-shot-RAG ontology-driven prompting. Retrieval-augmented generation (RAG) conditions a model's output on evidence fetched per query, where a retriever scores the query against an embedding store and supplies the top-k items as context [11]. Here, the retrieved items are labelled correspondences drawn from LogMap's own anchor mappings (i.e., high-confidence output the matcher produces without requiring any curated examples). Thus, the method is self-supervised and deployable in settings where no (partial) reference alignment exists. The top-k results fill the few-shot slot in Figure 1, such that half are kept as positive examples, and half are corrupted into negatives by replacing the target with a sibling. This comes with the cost of maintaining and extending a task-agnostic embedding store of mappings (i.e., for each new mapping task, new positive and negative examples can be added to the store). The use of LogMap's anchor mappings as source of positive and negative mappings avoid any potential leakage.

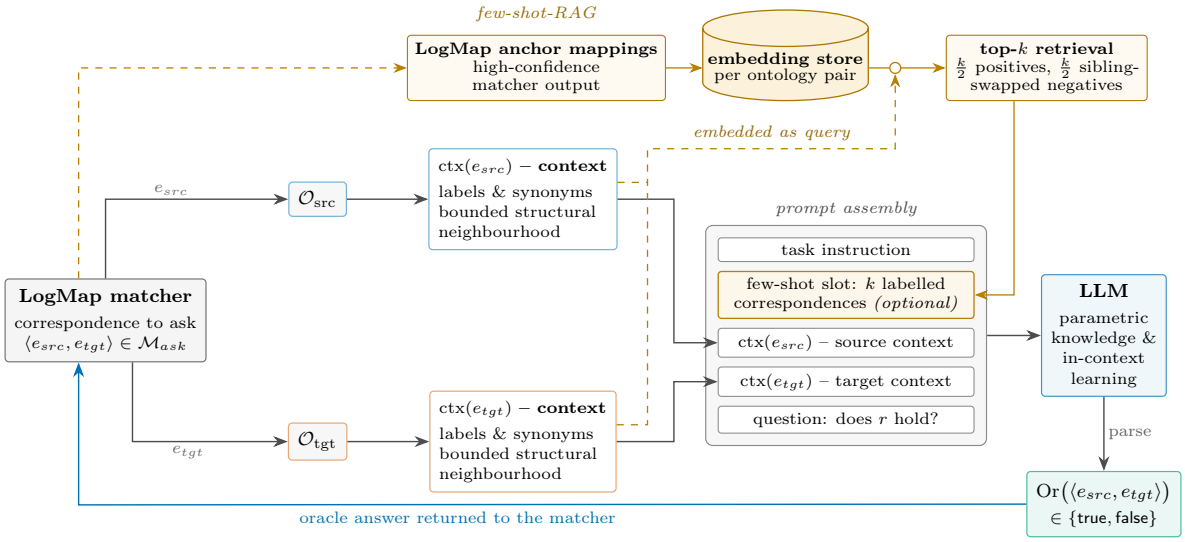


Figure 1: An illustration of ontology-driven prompting. The correspondence $\langle e_{src}, e_{tgt} \rangle$ selects an entity in each ontology, whose labels, synonyms, and bounded structural neighbourhood are provided by $\mathrm{ctx}(\cdot)$ as text. Prompt assembly concatenates a task instruction, an optional few-shot slot of k labelled correspondences, drawn from LogMap's anchors, the two textual contexts, and the question of whether r (i.e., equivalence) holds. The model's completion $\{\mathrm{true}, \mathrm{false}\}$ is returned to the matcher as $\mathrm{Or}(\langle e_{src}, e_{tgt} \rangle)$.

4. Evaluation on OAEI Tasks

We evaluate eight models chosen to span capability, cost, and openness: one capable API-only model (Gemini 2.5 Flash), a cost-effective open-weight model (DeepSeek V4 Flash), and several large and medium-sized self-hosted options (Qwen3.5 122B, Qwen3 235B, Mistral Small 3.2 24B, Gemma 3 27B); we other models included within on-going work (Llama 3.1 70B, Qwen3 8B).[3] Every run fixes a single prompt template with reasoning disabled and temperature 0, maximising determinism and cross-model comparability. We use SapBERT [12] for retrieval in HUMAN–MOUSE and Bio-ML, whereas `all-MiniLM-L12-v2` [13] is applied for Knowledge Graph (KG). The oracle adjudicates $|\mathcal{M}_{ask}| = 259$ (HUMAN–MOUSE), 4, 103 (Bio-ML), and 686 (KG) candidate pairs, retrieving from anchor pools of 1, 145, 17, 910, and 20, 507, respectively. The mean prompt length is 225 tokens at k=0, rising to 504 and 783 at k=2 and k=4. The LogMap *baseline* row in Table 1 labels a candidate positive if it occurs in the final alignment and negative otherwise; it provides the floor, upon which we must improve.

Table 1 reports sensitivity, specificity, and J on $\mathcal{M}_{ask}$ across the example sweep $k \in \{0, 2, 4\}$. The two matching regimes behave very differently at zero-shot. On HUMAN–MOUSE, every oracle is already sensitivity-saturated ($\sim$ 0.94–0.99), so there is little room for retrieval to add true positives. On KG the position is reversed: oracles are conservative, with high specificity but low sensitivity, and this is precisely where retrieval helps most. Adding two examples lifts Mistral Small's KG sensitivity from 0.673 to 0.935 and Qwen3 235B's from 0.577 to 0.843, which drives the track's large J improvements. The best KG oracle improves from J=0.719 at k=0 to 0.841 at k=2, then eases to 0.830 at k=4. For the larger models on HUMAN–MOUSE and Bio-ML, k=2 is typically near-optimal and k=4 adds token cost without benefit or mildly hurts. For example, DeepSeek V4 Flash on HUMAN–MOUSE moves 0.651 $\rightarrow$ 0.675 $\rightarrow$ 0.620. Importantly, trends are not uniformly monotone. For instance, Mistral Small degrades with k on HUMAN–MOUSE yet posts the best KG score at k=2, ceding the k=4 lead to Gemini 2.5 Flash.

On KG specifically, six of the eight models improve in J from k=2 to k=4 (e.g., Gemini 2.5 Flash 0.754 $\rightarrow$ 0.830, Llama 3.1 70B 0.748 $\rightarrow$ 0.810), with only Mistral Small and Qwen3 235B peaking at k=2. Gemma 3 27B and Qwen3 8B dip below their zero-shot J at k=2 but recover by k=4. Overall,

[3] A broader sweep over additional models is available as a companion resource at https://city-artificial-intelligence.github.io/logmap-llm-exp/.

Table 1
Sensitivity, specificity, and Youden's J on $\mathcal{M}_{ask}$ (primary) with final-alignment F_1 (secondary), across $k \in \{0, 2, 4\}$. KG F_1 is measured using its partial reference alignment semantics. "†" marks a score impacted by a non-zero call-failure rate. Best J per track within each k block in **bold**.

	HUMAN–MOUSE				Bio-ML				Knowledge Graph			
Model	sens.	spec.	J	F_1	sens.	spec.	J	F_1	sens.	spec.	J	F_1
LogMap *(baseline)*	1.000	0.000	0.000	0.881	0.827	0.173	0.000	0.700	0.142	0.960	0.092	0.781
Zero-shot (k=0)												
DeepSeek V4 Flash	0.970	0.681	**0.651**	0.899	0.949	0.543	0.492†	0.717	0.610	0.876	0.581	0.782
Gemini 2.5 Flash	0.970	0.596	0.565	0.896	0.930	0.547	0.477	0.716	0.720	0.807	0.604	0.783
Mistral Small 3.2 24B	0.988	0.628	0.616	0.898	0.929	0.498	0.428	0.715	0.673	0.976	0.644	0.783
Qwen3 235B	0.939	0.660	0.599	0.896	0.884	0.631	**0.515**	0.717	0.577	0.988	0.562	0.783
Qwen3.5 122B	0.976	0.649	0.625	0.898	0.927	0.573	0.500	0.718	0.736	0.976	0.707	0.784
Llama 3.1 70B	0.970	0.628	0.597	0.897	0.944	0.493	0.438	0.715	0.850	0.855	0.669	0.783
Gemma 3 27B	0.988	0.351	0.339	0.890	0.969	0.268	0.237	0.709	0.930	0.732	**0.719**	0.784
Qwen3 8B	0.964	0.436	0.400	0.891	0.933	0.410	0.342	0.712	0.504	0.982	0.482	0.783
Few-shot-RAG (k=2)												
DeepSeek V4 Flash	0.952	0.723	**0.675**	0.899	0.914	0.614	**0.527**†	0.717	0.770	0.948	0.706†	0.783
Gemini 2.5 Flash	0.964	0.596	0.559	0.896	0.926	0.557	0.483†	0.717	0.826	0.942	0.754	0.784
Mistral Small 3.2 24B	0.982	0.564	0.546	0.896	0.956	0.387	0.343	0.712	0.935	0.925	**0.841**	0.784
Qwen3 235B	0.964	0.638	0.602	0.897	0.923	0.506	0.429	0.715	0.843	0.942	0.771	0.784
Qwen3.5 122B	0.958	0.681	0.638	0.898	0.918	0.595	0.514	0.718	0.760	0.936	0.681	0.784
Llama 3.1 70B	0.945	0.628	0.573	0.896	0.924	0.550	0.474	0.716	0.885	0.891	0.748	0.784
Gemma 3 27B	0.970	0.500	0.470	0.893	0.918	0.423	0.341	0.712	0.821	0.815	0.590	0.784
Qwen3 8B	0.939	0.585	0.525	0.894	0.888	0.514	0.401	0.714	0.469	0.954	0.411	0.782
Few-shot-RAG (k=4)												
DeepSeek V4 Flash	0.939	0.681	0.620	0.897	0.919	0.618	**0.537**†	0.718	0.787	0.942	0.715	0.783
Gemini 2.5 Flash	0.970	0.596	0.565	0.896	0.931	0.548	0.479	0.717	0.902	0.942	**0.830**	0.784
Mistral Small 3.2 24B	0.982	0.532	0.514	0.895	0.957	0.367	0.324	0.712	0.946	0.885	0.802	0.784
Qwen3 235B	0.982	0.585	0.567	0.897	0.941	0.449	0.390	0.714	0.824	0.948	0.759	0.783
Qwen3.5 122B	0.958	0.670	**0.628**	0.898	0.915	0.596	0.511	0.718	0.768	0.942	0.696	0.784
Llama 3.1 70B	0.964	0.585	0.549	0.895	0.926	0.541	0.468	0.716	0.911	0.919	0.810	0.784
Gemma 3 27B	0.970	0.468	0.438	0.892	0.926	0.390	0.316	0.712	0.963	0.849	0.775	0.784
Qwen3 8B	0.915	0.670	0.585	0.895	0.881	0.529	0.410	0.714	0.636	0.948	0.571	0.783

retrieved anchors chiefly help the oracle accept true correspondences it would otherwise reject, and help most where zero-shot conservatism is the limiting factor. The oracle-level behaviour leaves downstream F_1 almost flat, since LogMap escalates only the correspondences it cannot already resolve. Since the matcher is already quite competent, the confident majority of its alignment is never re-examined.

5. Conclusion & Future Work

We introduced few-shot-RAG ontology-driven prompting, extending our prior work to include demonstrative context. Few-shot examples appear most useful where the zero-shot oracle fails conservatively. For instance, on knowledge graph, where sensitivity rather than specificity remains constrained, retrieved anchor-based examples deliver improvements that persist to k=4 for most models and lifts small open-weight models to competitive discriminative quality. In cases where oracles are already sensitivity-saturated, the effect is not as pronounced, levels off (Bio-ML), or mildly degrades (HUMAN–MOUSE) (at k=4). Including few-shot-RAG does not therefore yield uniform benefit, and whether the approach remains viable and token-efficient appears to depend on the LLM oracle, the task itself, and the prompt configuration.

Given the size of the configuration space, a natural next step, which we pursue in ongoing work, is a heuristic or learnt selector over model × prompt × few-shot configurations. Limitations of this work include the use of single runs per configuration, and the assumption that a negative may be reliably obtained through sibling-substitution.

Declaration on Generative AI

During the preparation of this work, the authors used Grammarly to check grammar, spelling, and readability; and Google Search with Gemini to suggest alternative phraseology. After using these tools, the authors reviewed and edited the content as needed to take full responsibility for the publication's content.

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